Stat 451 Machine Learning Report

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1 Introduction

Within our machine learning project, we analyzed a dataset containing the top 1,000 ranked YouTube videos, using variables: likes, views, category, and year published. To explore our question, "Is rank a meaningful metric for content creators to consider when planning videos?", we built models aimed at identifying which characteristics most strongly influence a video's overall rank. We experimented with multiple techniques, including PCA, k-NN, Multiple Linear Regression, and ensemble methods, Gradient Boosting and Random Forests. Although no model achieved particularly high accuracy, the Random Forest model demonstrated the strongest predictive capability, suggesting that the number of likes has the greatest impact on video rank. Ultimately, we concluded that YouTube is highly subjective and rank is likely calculated on features not represented in our data, therefore making it difficult to predict how well a video will perform.

Our dataset, from Kaggle, has 1000 rows with 7 variables. To clean the data, we removed the dislikes column due to hundreds of missing values and upon discovering YouTube creators can hide/manipulate their dislike count. Category had a few missing variables, but we filled them with 'Not Specified' and used one-hot encoding. It is important to note that this dataset represents the top 1,000 ranked YouTube videos as of Jan 27th 2025, meaning our analysis reflects only highly popular content rather than the broader population of all videos on the platform. This limits generalizability, but provides insight into what distinguishes top-performing content.

2 Models

We used multiple different modeling methods.

2.1 Multiple Linear Regression

First, we chose Multiple Linear Regression with a log based scale, in which we used video views, likes, published and category as our X and rank as our y.

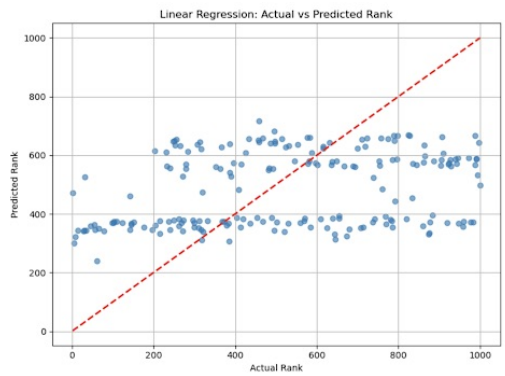

Figure 1: Multiple Linear Regression

The graph above shows the results we received. There was not strong predictive behavior, R2 score was 0.06.

2.2 k-NN

Then we explored k-NN using likes and views (X) to predict rank (y) which was also poor. The error rate was 318.4 and had negative accuracy scores.

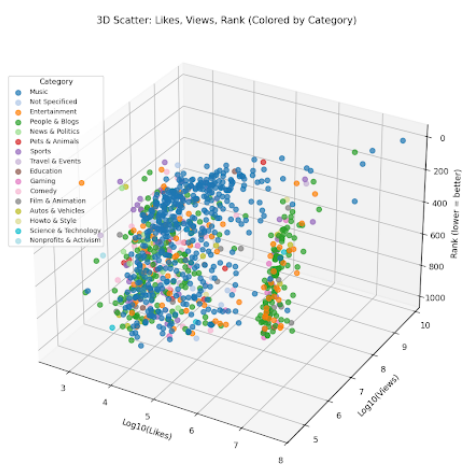

Figure 2: kNN

The 3D scatter plot above shows likes and views are poor at predicting rank.

2.3 PCA

Additionally, we worked with PCA. Although it had a good model fit, 95 percent explained variance with two components, it is not a predictive model.

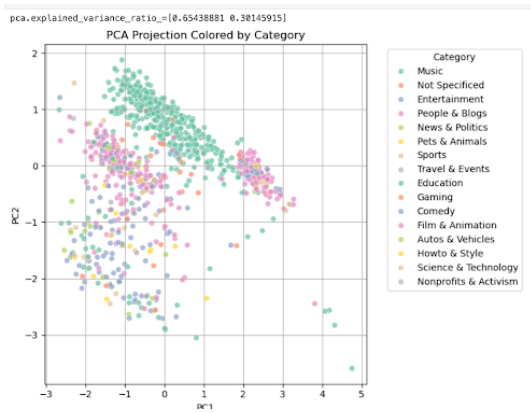

Figure 3: PCA

The graph did tell us high-ranking videos have similar feature patterns.

2.4 RandomForest

Finally, we tried Gradient Boosting and Random Forest with X features, published, likes, video views and category, and rank as y. They were both poor predictive models. Gradient Boosting received a 0.47 accuracy score and Random Forest received 0.53 validation score.

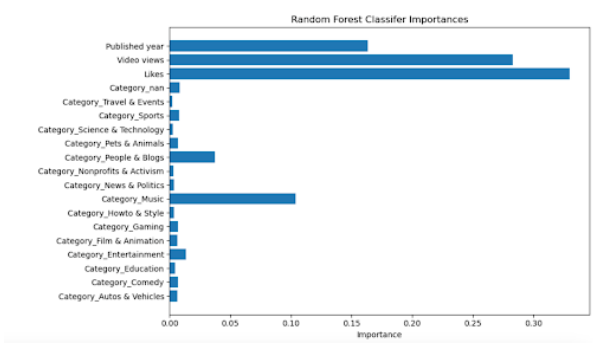

Figure 4: Random Forest

Likes, video views and music category are highest importance in predicting rank.

3 Conclusion

Overall, our various models showed that YouTube rank is very difficult to predict with the selected features in our data set. These results show that other key features, such as retention time, click rate, watch time, etc. are missing from the data. YouTube's ranking algorithm relies on a complex combination of engagement signals and behavioral metrics, many of which are not publicly disclosed and clearly not reflected in our dataset. Future work could expand the dataset to include the key feature engagement metrics which are more representative of YouTube's algorithm. Additionally, exploring larger datasets across multiple years could provide stronger generalization and reveal evolving trends in ranking behavior.

4 Contributions

	Proposal	Coding	Presentation	Report
Anna	1	1	1	1
Gwendolyn	1	1	1	1
Emma	0	1	1	1
Sofia	0.2	1	1	1
Nanddika	0	1	1	1